1. Clinical Question & Core Hypothesis

Clinical Question

BCG efficacy depends on local immune priming, so why do some tumors with evident local immune structures fail to respond to BCG? The KG suggests a chain where ERG (TMPRSS2/ERG) cooperates with PI3K activation to upregulate PD-L1, potentially creating an immunosuppressive niche that undermines BCG-driven immunity. 2) Tier 1 paths used: (a) TMPRSS2/ERG → cooperativity with PI3-kinase pathway activation → PD-L1 upregulation (mechanistic backbone explaining tumor-intrinsic immune suppression), and (b) BCG → upregulation of PD-L1 on target cells (links therapy to PD-L1 dynamics). I built on these because they directly explain how tumor signaling can neutralize local immune stimulation. 3) Tier 2 evidence supporting the claim: PSMA enrichment in cancer stem cells (links a resistant subpopulation that may persist despite immune activation), androgen-deprivation/prostate context (therapy-driven selection pressures that can alter signaling), and the general node that PD-L1 expression on cancer cells mediates immune evasion—these reinforce why PD-L1 elevation would blunt BCG benefit. 4) Why I chose therapy_response as the endpoint: the KG centers on BCG immunotherapy and PD-L1 modulation, so a therapy-response proxy (BCG response) is the most relevant, testable clinical outcome. 5) What I deliberately EXCLUDED and why: I excluded androgen-deprivation and PSMA as anchor genes to keep the mechanistic hypothesis sharp (max 2 anchors); they remain as secondary modifiers for follow-up. I also excluded broad pathway components (e.g., AKT, mTOR) from the core claim to avoid diluting the central ERG→PI3K→PD-L1 mechanism?

Working Hypothesis

In TLS-high breast cancer tumors, ERG-driven activation of the PI3K pathway induces upregulation of PD-L1 on tumor and/or immune cells within TLS structures. This PD-L1 upregulation creates an immunosuppressive TLS microenvironment that attenuates TLS-mediated immune priming and reduces response to BCG-like immunotherapies. Consequently, TLS-high tumors that are ERG-positive and PD-L1-high will show poorer therapy response compared with TLS-high, ERG-negative or PD-L1-low tumors. This effect may be modulated by additional Tier-2 modifiers (e.g., PIK3CA gain-of-function, PTEN loss, TMPRSS2 expression, PSMA expression). Importantly, inhibiting the PI3K pathway or blocking PD-L1 is predicted to restore TLS functionality and improve therapy response, providing a testable mechanism for combination strategies that re-engage TLS-driven immunity.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	ERG, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ERG: ERG → cooperativity with PI3-kinase pathway activation → PD-L1 upregulation (mechanistic backbone explaining tumor-intrinsic immune suppression), and (b) BCG → upregulation of PD-L1 on target cells (links therapy to PD-L1 dynamics).

PD-L1: PD-L1 upregulation (mechanistic backbone explaining tumor-intrinsic immune suppression), and (b) BCG → upregulation of PD-L1 on target cells (links therapy to PD-L1 dynamics).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PSMA	Supporting modifier from knowledge graph
Secondary	TMPRSS2	Supporting modifier from knowledge graph
Secondary	PIK3CA	Supporting modifier from knowledge graph
Secondary	PTEN	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: TCGA-BRCA integrative analysis for TLS_score, ERG/PD-L1 status, and PI3K pathway markers;

Prediction 2: in vitro manipulation of ERG/PI3K/PD-L1 in breast cancer cell lines with TLS-like co-culture assays;

Prediction 3: organoid or humanized mouse models to evaluate TLS formation and response to checkpoint blockade and PI3K inhibitors, with imaging readouts for TLS integrity and PD-L1 expression.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- PSMA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TMPRSS2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.